Tzu-Hsuan Chou, Ph.D.

Qualcomm Technologies, Inc., San Diego, CA, USA
tzuhsuan.henry.chou@gmail.com

EDUCATION

2017 – 2022 **Ph.D. in Electrical and Computer Engineering** **Purdue University**
Dissertation: Channel Training and Signal Processing for Massive MIMO Wireless Communications (co-advisors: David J. Love and Nicolo Michelusi)

2011 – 2013 **M.S. in Electrical and Control Engineering** **National Chiao Tung University**
Thesis: Feedback for Time-correlated MIMO-OFDM System using Predictive Quantization of Bit Allocation and Subcarrier Clustering (advisor: Y.-P. Lin)

2007 – 2011 **B.S. in Electrical and Computer Engineering** **National Chiao Tung University**

RESEARCH FIELDS OF INTEREST

Next-Gen Wireless Communications, MIMO, Signal Processing, mmWave/Sub-THz, 5G/6G and Wi-Fi

PROFESSIONAL EXPERIENCE

2022 – Present **Wireless Systems Engineer**
Qualcomm Technologies, Inc., CA, USA

2017 – 2022 **Graduate Research Assistant**
Department of Electrical and Computer Engineering, Purdue University, IN, USA

2014 – 2017 **4G LTE Software Engineer**
MediaTek, Inc., Hsinchu, Taiwan

2011 – 2013 **Graduate Research Assistant**
Department of Electrical and Control Engineering, National Chiao Tung University, Taiwan

PUBLICATIONS

Journal Articles

1. **T.-H. Chou**, C.-H. Liu, J. Jiang, "Distribution-Aware GMD Transceiver Design for Probabilistic Shaping in MIMO," *IEEE Trans. Veh. Technol.*, early access, 2026.
2. M. K. K. Gowda, **T.-H. Chou**, B. Lee, N. Michelusi, D. J. Love, Y. Zhang, J. V. Krogmeier, "Distributed Machine Learning for Low-Latency Localization in Cell-Free Massive MIMO Systems," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 19643–19659, 2026.
3. **T.-H. Chou**, N. Michelusi, D. J. Love, J. V. Krogmeier, "Joint UAV Placement and Transceiver Design in Multi-User Wireless Relay Networks," *IEEE Trans. Commun.*, vol. 73, no. 11, pp. 12820–12835, Nov. 2025.
4. **T.-H. Chou**, N. Michelusi, D. J. Love, J. V. Krogmeier, "Compressed Training for Dual-Wideband Time-Varying Sub-Terahertz Massive MIMO," *IEEE Trans. Commun.*, vol. 71, no. 6, pp. 3559–3575, Jun. 2023.
5. **T.-H. Chou**, N. Michelusi, D. J. Love, J. V. Krogmeier, "Fast Position-Aided MIMO Beam Training via Noisy Tensor Completion," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 774–788, 2021.

Conference Papers

1. **T.-H. Chou**, C.-H. Liu, B. Lee, J. Jiang, "EVM-Aware MIMO Transceiver Design for Wireless Communications," in *Proc. 60th Asilomar Conf. Signals, Syst., Comput.*, 2026, *accepted, to appear*.
2. M. K. K. Gowda, **T.-H. Chou**, B. Lee, N. Michelusi, D. J. Love, J. V. Krogmeier, "Hybrid Fingerprint-based Positioning in Cell-Free Massive MIMO Systems," in *Proc. 59th Asilomar Conf. Signals, Syst., Comput.*, Oct. 2025, pp. 525–529.
3. **T.-H. Chou**, N. Michelusi, D. J. Love, J. V. Krogmeier, "Wideband Millimeter-Wave Massive MIMO Channel Training via Compressed Sensing," in *Proc. IEEE Glob. Commun. Conf.*, 2021, pp. 1–6.
4. **T.-H. Chou**, N. Michelusi, D. J. Love, J. V. Krogmeier, "Millimeter Wave Beam Recommendation via Tensor Completion," in *Proc. IEEE Int. Conf. Commun.*, 2020, pp. 1–6.
5. Y.-P. Lin, **T.-H. Chou**, S.-M. Phoong, "Feedback for Time-Correlated MIMO-OFDM System Using Predictive Quantization of Bit Loading and Subcarrier Clustering," *IEEE International Conference on Digital Signal Processing (DSP)*, pp. 384–388, 2015.

PATENTS

***1 U.S. patent granted**; **1 U.S. patent application allowed**; **120 patent applications pending** within 4 years at Qualcomm; Physical-layer design in 6G and Wi-Fi — multi-antenna precoding, waveform, signaling, and transceiver design.*

1. **T.-H. Chou**, J. Jiang, J. Sun, Y. Zhang, "Shortened Time Domain Precoding Filters for Multi-Antenna Precoding," U.S. Patent Application US 2025/0070830 A1, published Feb. 2025.
2. L. Yang, **T.-H. Chou**, B. Tian, "Modulation of Extended Long Range Wireless Packets," U.S. Patent US 12,413,339 B2, granted Sept. 2025.

STANDARDS CONTRIBUTIONS

IEEE 802.11 (Wi-Fi)

*Contributions under Qualcomm affiliation to IEEE 802.11 Task Groups **bn (UHR / Wi-Fi 8)**, **bp (Ambient Power)**, and **bq (Integrated-MmWave)**. Available at mentor.ieee.org/802.11/documents. Author name also appears as "Henry Chou".*

1. L. Yang, **T.-H. Chou**, M. Rezk, B. Tian, Y. Kim, "M-LTF and SIG Field PAPR Reduction for 640MHz," *IEEE 802.11 TGbq (IMMW)*, doc. IEEE 802.11-26/1308, Jul. 2026.
2. L. Yang, **T.-H. Chou**, B. Tian, Y. Kim, M. Rezk, "Discussion on Beam Establishment and Channel Access," *IEEE 802.11 TGbq (IMMW)*, doc. IEEE 802.11-26/1302, Jul. 2026.
3. A. Chen, **T.-H. Chou**, B. Tian, S. Shellhammer, L. Yang, "Continued Discussion on Excitation Waveform for Bistatic Backscatter," *IEEE 802.11 TGbp (AMP)*, doc. IEEE 802.11-26/1009, May 2026.
4. L. Yang, **T.-H. Chou**, M. Rezk, B. Tian, Y. Kim, "Preamble Design for IMMW PPDU," *IEEE 802.11 TGbq (IMMW)*, doc. IEEE 802.11-26/0836, May 2026.
5. A. Chen, **T.-H. Chou**, B. Tian, S. Shellhammer, L. Yang, "Discussion on Excitation Waveform for Bistatic Backscatter — Follow Up," *IEEE 802.11 TGbp (AMP)*, doc. IEEE 802.11-26/0821, Apr. 2026.
6. A. Chen, **T.-H. Chou**, B. Tian, S. Shellhammer, L. Yang, "Discussion on Excitation Waveform for Bistatic Backscatter," *IEEE 802.11 TGbp (AMP)*, doc. IEEE 802.11-26/0798, Apr. 2026.
7. L. Yang, **T.-H. Chou**, B. Tian, Y. Kim, M. Rezk, "High Level Thoughts on IMMW PPDU Format," *IEEE 802.11 TGbq (IMMW)*, doc. IEEE 802.11-26/0471, Mar. 2026.
8. L. Yang, **T.-H. Chou**, B. Tian, Y. Kim, "High Level Thoughts on BW Modes and Numerology in IMMW," *IEEE 802.11 TGbq (IMMW)*, doc. IEEE 802.11-25/1737, Nov. 2025.

9. L. Yang, *et al.* (incl. **T.-H. Chou**), "PDT PHY Enhanced Long Range (ELR)," *IEEE 802.11 TGbn (UHR)*, doc. IEEE 802.11-24/1981, Nov. 2024.
10. L. Yang, **T.-H. Chou**, B. Tian, Y. Kim, "Enhanced Long Range (ELR) PPDU Design," *IEEE 802.11 TGbn (UHR)*, doc. IEEE 802.11-24/1478, Sept. 2024.

3GPP RAN1 (6G)

*Contributions under Qualcomm affiliation to **3GPP RAN1**; 3GPP tdocs are under company affiliation without individual authorship attribution. Physical-layer design in MIMO transceiver for uplink transmission, and modulation waveform.*

1. Qualcomm, "Modulation, Joint Channel Coding and Modulation," *3GPP TSG RAN WG1 **#126***, tdoc R1-2606911, Aug. 2026.
2. Qualcomm, "Uplink Transmission Schemes for Uplink Channels," *3GPP TSG RAN WG1 **#124/#125/#126***, tdocs R1-2601278/R1-2604698/R1-2606380, Feb.–Aug. 2026.

TEACHING / MENTORING EXPERIENCE

Teaching Assistant

2012 – 2013 **Calculus** *(undergraduate level)* National Chiao Tung University

Research Mentorship

2023 – 2025 Mentored graduate student — Originated research direction and provided technical guidance to junior Ph.D. student, resulting in **journal article (TWC)** and **conference paper (Asilomar)**.

PROFESSIONAL SERVICE EXPERIENCE

2026 – 2027 **Technical Program Committee Member**
IEEE International Conference on Communications (ICC 2027) — SPC Symposium

Peer Review (35 manuscripts reviewed across 12 venues)

IEEE *Journal on Selected Areas in Communications*
IEEE *Journal of Selected Topics in Signal Processing*
IEEE *Transactions on Communications*
IEEE *Transactions on Wireless Communications*
IEEE *Transactions on Vehicular Technology*
IEEE *Transactions on Network and Service Management*
IEEE *Communications Letters*
IEEE *Wireless Communications Letters*
IEEE *Signal Processing Letters*
IEEE *Access*
IEEE *Global Communications Conference (GLOBECOM)*
Elsevier *Digital Signal Processing*

HONORS AND AWARDS

2017 **ECE Fellowship for Ph.D. Student**
Purdue University, IN, USA

PROFESSIONAL MEMBERSHIP

2020 – Present **IEEE Member**
Institute of Electrical and Electronics Engineers (IEEE)